

Bambinos Marketing & Cohort Intelligence Platform

PRODUCTION ROLLOUT & EXECUTION PLAN: DIRECT META MARKETING INGESTION

Target Standard: v14 Production Live Standard	Scheduled Run: Daily 08:00 AM IST (02:30 UTC)
Database Protection: 500-Row Chunks + 50ms Throttle	Deployment Target: GCP Cloud Run Jobs / GitHub Actions

1. Executive Rollout Overview

This execution plan details the 5-phase rollout strategy to operationalize the Direct Meta Marketing Graph API pipeline into Bambinos' live database infrastructure. It enforces zero database CPU spikes, complete data privacy, zero-discrepancy financial reconciliation, and automated sub-50ms dashboard query serving.

2. Detailed 5-Phase Execution Sequence

Phase 1: Environment & Staging Schema Provisioning

- **Step 1.1 (Staging DDL Execution):** Execute queries/01_production/00_create_staging_tables.sql to create raw_meta_ads_delivery (clustered primary key) and data_pipeline_runs.
- **Step 1.2 (Credential Configuration):** Add META_ACCESS_TOKEN (System User Token with ads_read) and META_AD_ACCOUNT_IDS to backend/.env and Secret Manager.

Phase 2: Live Ingestion & Historical Backfill

- **Step 2.1 (Dry-Run Verification):** Run python backend/scripts/ingest_meta_direct.py --days 7 --dry-run to test API connectivity and action unnesting without DB writes.
- **Step 2.2 (30-Day Initial Backfill):** Execute live extraction: python backend/scripts/ingest_meta_direct.py --days 30 with 500-row batching and 50ms micro-throttles.
- **Step 2.3 (Staging Validation):** Verify row counts in raw_meta_ads_delivery and confirm SUCCESS status in data_pipeline_runs.

Phase 3: Materialized Cache Synchronization & Benchmarking

- **Step 3.1 (Stored Procedure Compilation):** Ensure procedures from 28_nightly_cache.sql (sp_refresh_cohort_cache_nightly) are compiled on MariaDB.
- **Step 3.2 (Rolling Cache Sync):** Run python backend/scripts/sync_cohort_cache.py --mode incremental across the active 35-day maturation window to populate cohort_detail_cache.
- **Step 3.3 (Latency Benchmarking):** Verify FastAPI endpoint response times drop from >15s to <50ms.

Phase 4: Quality Assurance & Reconciliation Gate

- **Step 4.1 (Zero-Discrepancy Audit):** Run python backend/scripts/verify_reconciliation.py to confirm: (1) SUM(spend) in staging matches SUM(spend) in cohort cache exactly, and (2) Unmapped campaign spend is <2%.
- **Step 4.2 (RAM Cache Invalidation):** Flush backend memory caches via cache_manager.clear() to propagate fresh figures immediately to all cards.

Phase 5: Automated Cloud Orchestration & Monitoring

- **Option A (GitHub Actions):** Push repository with .github/workflows/daily-marketing-sync.yml scheduled for daily **08:00 AM IST (02:30 UTC)** with manual on-demand trigger support.
- **Option B (GCP Cloud Run Job):** Execute bash deploy_marketing_job.sh to create marketing-daily-sync Cloud Run Job in asia-south1 matching bambinos-core conventions.

3. Database Health & Zero-CPU-Spike Safeguards

Safeguard Mechanism	Engineering Implementation	Production Impact
Batch Size Cap	500 rows per transaction chunk	Prevents InnoDB buffer pool exhaustion and row-lock escalation.
Inter-Batch Throttling	50ms micro-sleep (time.sleep(0.05))	Yields CPU and lock resources to concurrent web queries and CRM users.
In-Memory Offloading	Actions unnested in Python RAM	Zero JSON parsing or regex evaluation on the database CPU.
Idempotent Upsert	ON DUPLICATE KEY UPDATE	Re-running backfills updates existing rows with zero duplicate spend.
Sliding Window Sync	Rolling 35 days (incremental mode)	Refreshes only active maturing cohorts rather than scanning years of history.

4. Quick-Reference Operational Commands

Operational Action	Terminal Command
Run Full Test Suite	python -m unittest discover -s backend/tests -p "test_*.py"
Test Dry-Run Master Pipeline	python backend/scripts/run_daily_pipeline.py --days 3 --dry-run
Execute Live 30-Day Backfill	python backend/scripts/run_daily_pipeline.py --days 30
Deploy Cloud Run Job	bash deploy_marketing_job.sh

Rollout Approval: This execution plan provides an end-to-end, zero-risk pathway to live production. All code, DDL, procedures, unit tests, and schedulers are verified and ready for execution upon credential input.